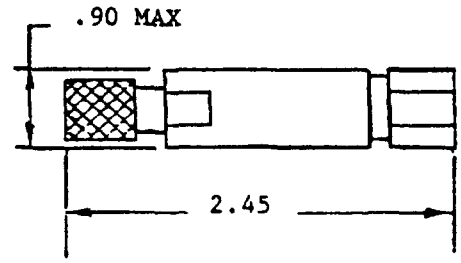
MILITARY SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY SERIES N (PLUGS AND JACKS,SEMIRIGID CABLE, MALE, CLASS 2) (FOR USE WITH MIL-C-28830/3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.



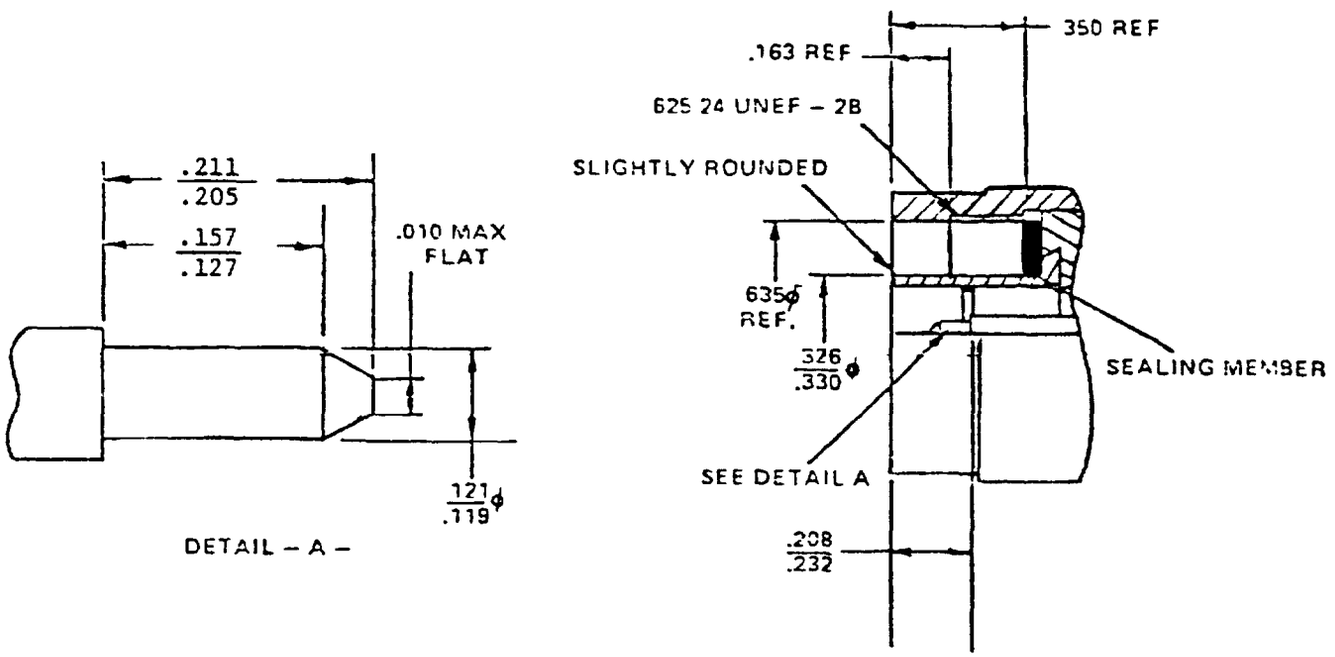
INCHES	MM
0.90	22.86
2.45	62.23

NOTES:
1. Dimensions are inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:16Z
Check the source to verify that this is the current version before use.



INCHES	MM
.127	3.23
.157	3.99
.163	4.14
.205	5.21
.208	5.28

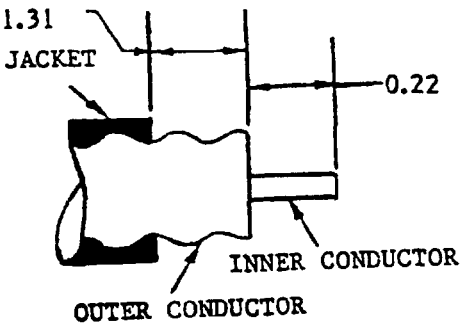
INCHES	MM
.211	5.36
.232	5.89
.328	8.33
.330	8.38
.350	8.89

INCHES	MM
.625	15.88
.635	16.13
.010	.254
.119	3.02
.121	3.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Mating dimensions for pin terminations - plugs.

Source: https://assist.dla.mil -- Downloaded: 202640-21T17:16Z
Check the source to verify that this is the current version before use.



INCHES	MM
0.22	5
1.31	33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Cable outer conductor to be cut at crest only.

FIGURE 3. Cable stripping dimensions for connectors.

Source: https://assist.dla.mil -- Downloaded: 202640-21T17:16Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Nominal impedance: 50 ohms
Frequency range: 0 to 8.1 GHz
Voltage rating: 500 Volts
Temperature range: -40°C to +100°C
Peak RF power: 5.0 kW

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Force to engage and disengage:
    Longitudinal force - Not Applicable
    Torque - 6 inch pounds maximum.
Coupling proof torque: 15 inch pounds minimum.
Inspection conditions: 6-10 inch pounds
Mating characteristics: See figure 2 for dimensions.
Hermetic seal: Not applicable.
Leakage (pressurized connectors): Not applicable.
Insulation resistance: Method 302, test condition B, MIL-STD-202.
    5,000 megohms minimum.

Center contact retention: 6 pounds minimum
    Radial torque: Not applicable.
Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.
Voltage standing wave ratio (VSWR) per Connector: 0.5 to 8.0 GHz 1.2:1

Swept frequency VSWR test setup:
    Item 6- VSWR shall be less than 1.008 + .002 f (f in GHz).
    Item 16 - VSWR shall be less than 1.008 + .002 f (f in GHz).
    Second step of VSWR checkout procedure - VSWR shall be less than
        1.012 + .005 f (f in GHz)
    Group B inspection - 1.024 + .007
    Qualification and group C inspection - shall not exceed 1.08

Connector durability: A maximum of 500 cycles at 12 cycles per minute

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:16Z
Check the source to verify that this is the current version before use.

Contact resistance: In milliohms maximum.

	Initial	After environment
Inner contact:	2.0	2.0
Outer contact:	0.3	0.3

Dielectric withstanding voltage at sea level: 2000 volts
Method 301 of MIL-STD-202

Vibration, high frequency: Method 204, test condition D, MIL-STD-202. Cable clamps shall be mounted on the vibration table 6 inches minimum from the mating face of the connector pair.

Shock: Method 213, test condition I, MIL-STD-202.

Temperature cycling: Method 102, test condition C, MIL-STD-202, except test high temperature shall be + 115°C.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be of at least 200 Megohms within 5 minutes after removal from humidity.

Corona level: Not applicable

RF high potential withstanding voltage: 1500 Volts RMS
Frequency - 5-7 MHz
Leakage current - Not applicable.

Cable retention force: 200 pounds minimum

Torque: 55 inch ounces

Coupling mechanism retention force: 100 pounds min

RF leakage : -90 dB Tested at a frequency of 2-3 GHz

RF insertion loss: dB max - 0.10 dB max at 4.0 GHz

Preparing Activity
NAVY-EC
(Project Number 5935-N169-1)

☆U.S. GOVERNMENT PRINTING OFFICE 1980 - 703 023/4581

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-39012/117 (EC)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:16Z
Check the source to verify that this is the current version before use.